Sample mean

$$\bar{x} = A + \frac{\sum_{i=1}^{10} u_i}{n}$$ where A is assumed mean

$= 85 + 0 = 85$

Sample standard deviation

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{10} u_i^2}$$

$$= \sqrt{\frac{1}{9} \times 690}$$

$$= \sqrt{76.67}$$

$= 8.756$

Hence,

$$t_0 = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{85-90}{8.756/\sqrt{10}} = \frac{-5}{2.77}$$

$= -1.806$ and

$|t_0| = 1.806$

Step 6 : Critical value

Since H_1 is two-sided alternative hypothesis, the critical value at $\alpha = 0.05$ is $t_e = t_{n-1, \frac{\alpha}{2}} = t_{9,0.025} = 2.262$

Step 7 : Decision

Since it is two-tailed test, elements of critical region are defined by the rejection rule $|t_0| > t_e = t_{n-1, \frac{\alpha}{2}} = t_{9,0.025} = 2.262$. For the given sample information $|t_0| = 1.806 < t_e = 2.262$. It indicates that given sample does not provide sufficient evidence to reject H_0. Hence, we conclude that the class average scores is 90.

2.1.5 Test of Hypotheses for Equality of Means of Two Normal Populations (Independent Random Samples)

Procedure:

Step 1 : Let μ_X and μ_Y be respectively the means of population-1 and population-2 under study. The variances of the population-1 and population-2 are assumed to be equal and unknown given by σ^2.

Frame the null hypothesis as $H_0 : \mu_X = \mu_Y$ and choose the suitable alternative hypothesis from (i) $H_1 : \mu_X \neq \mu_Y$ (ii) $H_1 : \mu_X > \mu_Y$ (iii) $H_1 : \mu_X < \mu_Y$

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